DO NOW — RECALL

CSS and JavaScript

Name: _____ Class: _____ Date: _____

1. What does the HTML id attribute identify?

2. What does a CSS selector do?

3. What does a button click trigger in JavaScript?

4. Binary search in [2,4,6,8,10,12,14] for 12: show midpoint indices.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A specific DOM element.

2. Model answer

Targets elements to style.

3. Model answer

A registered event handler.

4. Model answer

mid3→8; low4 high6 mid5→12, found indexes 3,5.

Vocabulary — know these terms

CSS — Style sheets controlling presentation.

selector — Targets HTML/DOM nodes.

DOM — Browser's document object model.

event handler — Code invoked by event.

textContent — Text stored in a DOM node.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Show method

Assigning textContent replaces displayed text with 'On'; it is safer for text than injecting untrusted HTML.

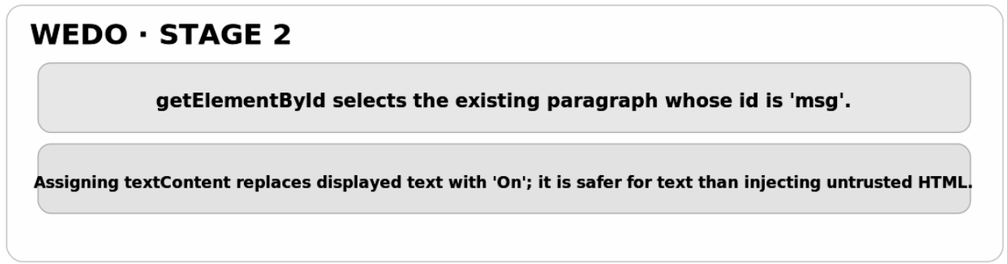
IMPROVE: Explain one correction to your own reasoning.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A page contains p id='msg' showing 'Off'. Click handler changes textContent to 'On'. Explain DOM result.

getElementById selects the existing paragraph whose id is 'msg'.



WEDO · STAGE 2: getElementById selects the existing paragraph whose id is 'msg'. then Assigning textContent replaces displayed text with 'On'; it is safer for text than injecting untrusted HTML.

NEW LEARNING

Read, annotate and explain

Understand

CSS controls presentation using selectors and declarations; `#status` targets one element by id.

Trace

JavaScript can react to browser events and modify the DOM. The handler does not execute until a click event occurs.

Evaluate

The document's structure remains HTML, styles remain CSS, and interactive behaviour is JavaScript. Validate incoming data on the server even if scripts check it first.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

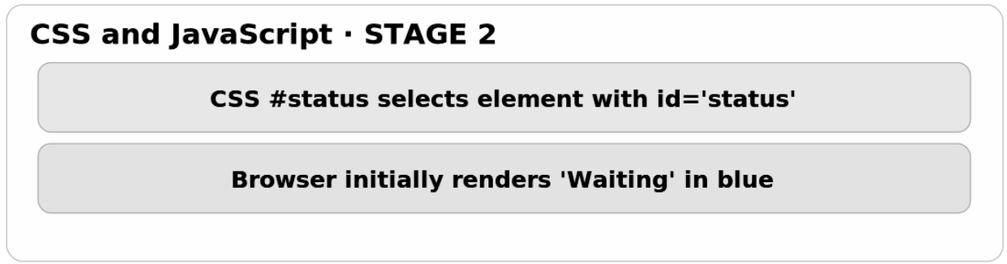
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



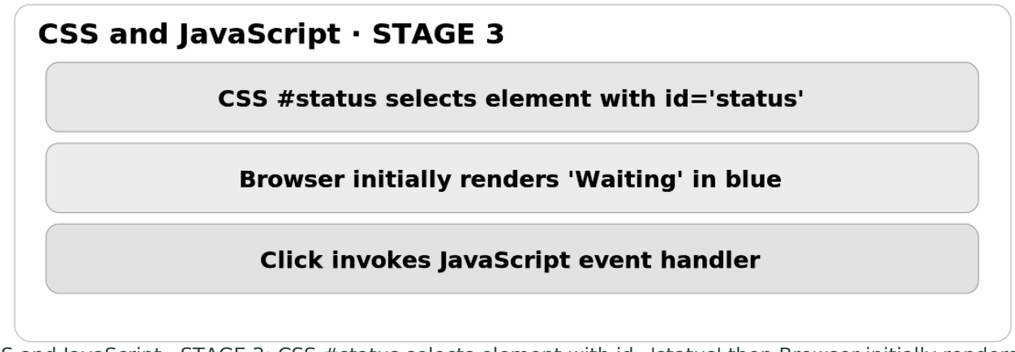
CSS and JavaScript · STAGE 1: CSS #status selects element with id='status'

New Learning



CSS and JavaScript · STAGE 2: CSS #status selects element with id='status' then Browser initially renders 'Waiting' in blue

New Learning



CSS and JavaScript · STAGE 3: CSS #status selects element with id='status' then Browser initially renders 'Waiting' in blue then Click invokes JavaScript event handler

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

JavaScript is needed to define every HTML heading and paragraph.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

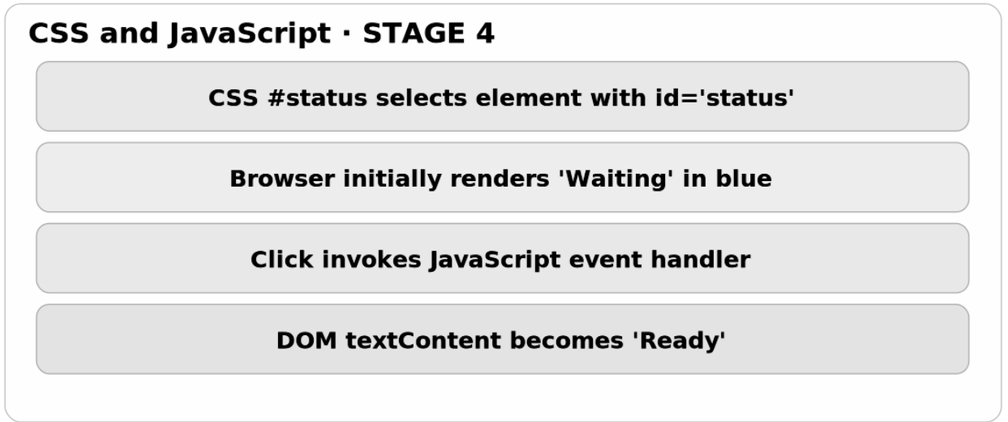
Question 3

Compare changing text with textContent and inserting untrusted input via innerHTML. [6]

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



CSS and JavaScript · STAGE 4: CSS #status selects element with id='status' then Browser initially renders 'Waiting' in blue then Click invokes JavaScript event handler then DOM textContent becomes 'Ready'

I DO — WORKED EXAMPLE

Follow the worked method

Trace click count initial 0 when handler increments by one on each click. Two clicks occur.

Predict first

Start with count=0; first click handler adds one making count=1.

Show method

Second click adds one again giving count=2; no change happens before an event.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

In the code, what does CSS #status apply and what is initial text? [4]

Question 2

Write JavaScript so a click on button id='go' changes p id='status' to 'Complete'. [6]

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first getElementById selects the existing paragraph whose id is 'msg'.

DISCUSS — Compare the representation or steps you used. Explain one decision and challenge one possible misconception.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace click count initial 0 when handler increments by one on each click. Two clicks occur.

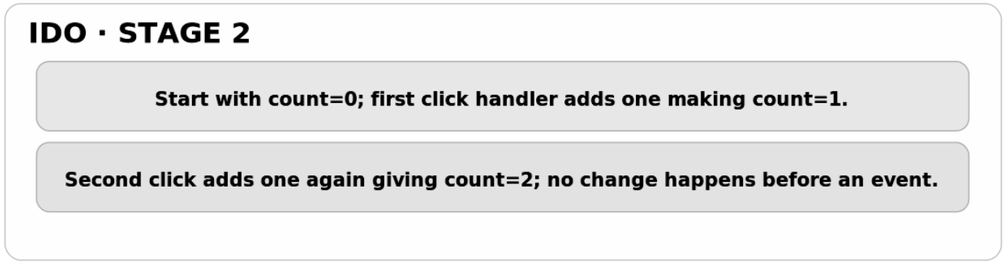
Start with count=0; first click handler adds one making count=1.



IDO · STAGE 1: Start with count=0; first click handler adds one making count=1.

Trace click count initial 0 when handler increments by one on each click. Two clicks occur.

Start with count=0; first click handler adds one making count=1.



IDO · STAGE 2: Start with count=0; first click handler adds one making count=1. then Second click adds one again giving count=2; no change happens before an event.

WE DO — GUIDED PRACTICE

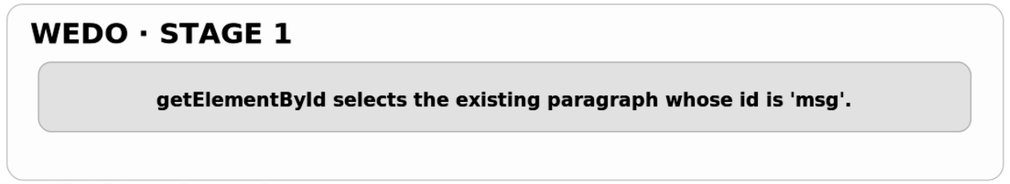
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A page contains p id='msg' showing 'Off'. Click handler changes textContent to 'On'. Explain DOM result.

getElementById selects the existing paragraph whose id is 'msg'.



WEDO · STAGE 1: getElementById selects the existing paragraph whose id is 'msg'.